

Pranav Nuti

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Employment

The University of Chicago Booth School of Business 2024–
Principal Researcher
Mentored by Rad Niazadeh

Education

Stanford University 2018–2024
PhD in Mathematics
Advised by Jan Vondrák

Indian Institute of Science (IISc), Bangalore 2014–2018
Bachelor of Science (Research), Gold Medalist in Mathematics

Research

Journal Publications and Revisions.....

Prophet Inequalities with Cancellation Costs Operations Research, Minor Revision
with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák

Toward an Optimal Contention Resolution Scheme for Matchings Mathematical Programming 2025
with Jan Vondrák

Online Matching and Contention Resolution for Edge Arrivals with Vanishing Probabilities Operations Research, Minor Revision
with Will Ma, Calum MacRury

The Secretary Problem with Distributions Mathematics of Operations Research, Major Revision

Conference Publications.....

Stationary Online Contention Resolution Schemes EC 2026
with Mohammad Reza Aminian and Rad Niazadeh

Optimal Prophet Inequalities with Buyback EC 2026
with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák

Static Pricing for Single Sample Multi-unit Prophet Inequalities SOSA 2026
with Peter Westbrook

Online Matching and Contention Resolution for Edge Arrivals with Vanishing Probabilities EC 2024
with Will Ma, Calum MacRury

Prophet Inequalities with Cancellation Costs STOC 2024
with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák

Secretary Problems: The Power of a Single Sample SODA 2023
with Jan Vondrák

A Tight Competitive Ratio for Online Submodular Welfare Maximization ESA 2023
with Amit Ganz and Roy Schwartz

Toward an Optimal Contention Resolution Scheme for Matchings <i>with Jan Vondrák</i>	IPCO 2023
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The Secretary Problem with Distributions	IPCO 2022
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Working Papers.....

Single-Sample Prophet Inequalities for Knapsack Problems <i>with Rebecca Reiffenhäuser and Alice Sayutina</i>	(submitted)
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Optimal Bayesian Online Allocation of Reusable Resources <i>with Mohammad Reza Aminian and Rad Niazadeh</i>	(submitted)
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Workshop Papers.....

Solving Open Problems in Operations Research Using AI <i>with Eric Fithian and Rad Niazadeh</i> Draft paper. Based on Open Problems in Operations Research. Also presented as a poster at Can AI do Theory? at STOC 2026.	AI-Driven Research in EconCS at EC 2026
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Bounded Best Buyers: Prophet Inequalities Revisited <i>with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák</i>	Easy Peasy at EC 2026
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Magicians Don't Move: An Easy Peasy OCRS <i>with Mohammad Reza Aminian and Rad Niazadeh</i>	Easy Peasy at EC 2026
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A Match Made by Entropy <i>with Mohammad Reza Aminian and Rad Niazadeh</i>	Easy Peasy at EC 2026
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Selected Talks.....

How to write and review papers <i>EC 2026 Mentoring Workshop</i>	2026
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Prophet Inequalities with Cancellation Costs <i>Chicago Junior Theorists Workshop</i>	2025
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What is a Good Model for Online Decision Making? <i>SURIM Talk, Stanford University</i>	2023
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Online Selection Problems and how Ramsey's Theorem Solves a Card Game <i>DACO Seminar, ETH Zürich</i>	2023
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Between the Secretary Problem and the Prophet Inequality <i>Data-Driven Decision Processes Talk, Simons Institute, UC Berkeley</i>	2022
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Maximum Matchings in Sparse Random Graphs <i>Department of Mathematics, Stanford University</i>	2022
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Teaching

Papers.....

Engaging Activities for Teaching Linear Algebra <i>with Shintaro Fushida-Hardy, Megan Selbach-Allen</i>	PRIMUS 2024
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Awards.....

Pólya Teaching Fellow Award <i>Awarded for outstanding teaching during the PhD program</i>	2024
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Robert Osserman Teaching Award <i>Awarded for outstanding contributions as a teaching assistant at Stanford University</i>	2022
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Hoefler Mentoring Award 2022
Awarded for supporting undergraduate writing at Stanford University

Courses Taught as Instructor.....

Linear Algebra, Stanford Summer Engineering Academy 2021, 2022

Equity and Inclusion Initiative, Stanford Engineering

This is a summer program that seeks to illuminate the brilliance of students who have been systemically marginalized in engineering. Along with teaching the course, in collaboration with my co-instructors, I also designed the curriculum. 38 classroom hours. Syllabus, lesson plans, handouts and descriptions of all activities are available.

Teaching Outside the University.....

Stanford Jail and Prison Education Project 2023

San Francisco County Jail # 5

The project provides education services to people who are incarcerated in Bay Area jails. I helped design the curriculum, and taught two seminars.

Undergraduate Student Mentorship.....

Stanford Undergraduate Research in Mathematics 2023

Department of Mathematics, Stanford University

The program is aimed at introducing Stanford undergraduate students to research in mathematics. I mentored Ethan Zhang, Ezra Steinberg, Peter Westbrook, and Quinn McIntyre. We proved several new results related to prophet inequalities. A report is available.

Directed Reading Program 2020–2022, 2024

Department of Mathematics, Stanford University

The program is aimed at helping undergraduates, especially members of historically underrepresented groups, participate in the larger mathematical community. I mentored several students: Alessandra Maranca, Sophia Sanchez, Ellen Xu, Donald Poindexter, Logan Bhamidipaty, and Sohan Vichare.

Graduate Student Mentorship.....

Stanford Summer Engineering Academy 2023

Equity and Inclusion Initiative, Stanford Engineering

I interviewed instructors to hire for the program, helped design a new curriculum, and provided feedback to instructors.

TA Mentoring Program 2022–2023

Department of Mathematics, Stanford University

The program is aimed at providing first-time TAs advice and feedback about leading section. I mentored Yosheb Getachew, Joao Campos Vargas, Zhenyuan Zhang, and Hongjian Yang.

Other Contributions

Organizer of the Student Probability Seminar 2021

Department of Mathematics, Stanford University

WINGS Representative 2019–2020

Department of Mathematics, Stanford University

The WINGS program aims to promote wellness amongst graduate students.

Other Awards

Bronze Medal, 54th International Mathematical Olympiad 2013

KVPY Fellowship 2013–2018

Awarded by the Govt. of India

A 100% scholarship for undergraduate education.

National Talent Search (NTS) Scholarship
Awarded by the Govt. of India to high school students

2010–2011